

Can a model learn to draw — one patch at a time?

Due: 6/5/2025 8AM

You will build a **causal patch transformer** — a model that generates images by predicting patches sequentially, left-to-right and top-to-bottom, using masked self-attention so each patch can only attend to patches that have already been generated. This is the image analogue of GPT: the same causal transformer architecture that generates text, applied to visual patch tokens.

You will then train a VAE on the same dataset and evaluate both models head-to-head across sample quality, training speed, and generation speed

AI tools policy: You may use Claude, Copilot, ChatGPT, or any coding assistant freely for code generation, debugging, and writing drafts. All significant uses must be documented in the AI usage log (see Deliverables). You will be evaluated on your ability to reason about and interpret what you built — not on whether you typed every line yourself.

What changes when AI writes your code

When the mechanical work is inexpensive, the assessment shifts from implementation to interpretation. The table below makes this mapping explicit.

Without AI tools (before)	With AI tools — what you are graded on
Implementing the causal mask correctly	Verifying no future-token leakage exists — explaining how you tested it and what would happen if it leaked
Writing the autoregressive sampling loop	Interpreting the attention weights at each generation step: what does the model look at when drawing patch t ?
Training a VAE baseline	Making a fair, evidence-based comparison across sample quality, training speed, and generation speed
Plotting training loss curves	Explaining a result from the VAE comparison that contradicted your expectation — and what it implies about the two model families

Connection to lecture material

This project is designed so that concepts from both lectures are load-bearing — neither can be removed without the system breaking.

From Lec 13 — Attention & Transformers	From Lec 14 — Generative AI
• Patch tokenization — images as token sequences • Sinusoidal positional encoding — generation order matters • Causal (masked) self-attention — no look-ahead permitted • Attention weight visualisation — interpret model focus during generation	• Autoregressive generation — explicit $p(x)$ via chain rule • Softmax over patch vocabulary — patch-level PixelCNN • VAE as baseline — implicit density, latent representation • Sample quality vs. training and generation speed trade-offs

Key distinction from a bidirectional model: Bidirectional (e.g. BERT-style) attention lets every token attend to all others — useful for understanding. Causal attention lets each token attend only to previous tokens — required for generation. Implementing and verifying this constraint correctly is the first technical challenge of this project.

System pipeline

The five stages below must all be running end-to-end before the investigation milestone begins.

① Tokenize image Patch extraction + pos. encoding	② Causal mask No future-token look-ahead	③ Transformer Predict next patch token	④ AR sampling Generate full image	⑤ VAE comparison Head-to-head analysis
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Milestones

1

Causal transformer — working implementation Days 1–3

Build a transformer trained to predict patch $t+1$ given patches $0\dots t$. The causal attention mask is the key structural element — implement it explicitly and include a test that verifies no future-token information leaks (e.g., ablate the mask and confirm performance degrades).

Train on CIFAR-10. First up-sample the image to 64×64 , use patch size of 8×8 , yielding 64 patches (tokens), plus the BOS token (beginning of sentence).

Deliverable: training loss curve across epochs + one sanity-check panel showing next-patch predictions for a held-out partial sequence.

AI tools note: expected approximately half a day of active work to scaffold the transformer and training loop

2

Autoregressive sampling and attention timelines Days 3–6

Implement the sampling loop: start from a [BOS] token, autoregressively predict and append patches until the image is complete. Generate a grid of 20 novel samples.

Then, for 5 of these generated images, capture and visualise the attention weights at each generation step. Produce two annotated attention-timeline figures that answer:

- Which previously-generated patches does the model attend to most strongly when drawing patch t ?
- Does the model develop local vs. global attention patterns across the sequence?
- Do early patches exert disproportionate influence over later ones?

Deliverable: 20-sample generation grid + 2 attention-timeline figures with written captions that interpret what is happening at each step.

AI tools note: the sampling loop is straightforward to generate with AI; the written interpretation of the attention timelines cannot be delegated

3

VAE baseline — head-to-head comparison Days 5–8

Train a VAE on the same dataset using the same number of training epochs. Evaluate both models on three axes:

- Sample quality — visual inspection grid, plus FID score or reconstruction MSE on the test set
- Training speed — wall-clock time per epoch at matched batch size
- Generation speed — seconds to produce one full image at inference

Write a 300-word comparison memo grounded entirely in your own results. Include explicit use-case recommendations: which model would you deploy for (a) fast prototyping, (b) maximum sample quality, and (c) a system that needs to compute $p(x)$ for anomaly detection? Justify each recommendation.

AI tools note: VAE scaffolding is fast; the three-axis comparison and use-case reasoning must be grounded in your own results and cannot be generated without running the experiments

4

Open investigation Days 7–11

Choose one of the questions in the next section. State a falsifiable hypothesis before running any experiments. Collect a minimum of four data points. Produce at least one results plot. Write a 400–600 word discussion.

You are expected to encounter at least one result that complicates or contradicts your hypothesis. When this happens, engage with it — do not re-run experiments until the result matches expectations. The quality of your reasoning about unexpected findings is the primary thing being assessed in this milestone.

Investigation questions — choose one

Select exactly one question. State your hypothesis in writing before running any experiments.

Investigation question	Description	Difficulty
Generation order	Compare raster order (left→right, top→bottom) vs. random order vs. center-out order. Which produces better samples? Does the attention pattern structure change with generation order? Explain why order might matter for a causal model.	Medium
Sampling temperature	Generate 20 samples at $\tau \in \{0.5, 0.7, 1.0, 1.3, 1.8\}$. Plot sample diversity (average pairwise MSE) and visual quality vs. temperature. Where is the sweet spot, and what does the shape of the curve reveal about how the model distributes probability mass?	Medium
Depth vs. width	Fix total parameter count (~2 M). Compare 2 deep vs. 8 shallow transformer layers. Do they produce different attention structures? Plot validation loss curves and sample grids side-by-side. Which architecture generalises better, and why?	Hard
Class-conditional generation	Prepend a learnable class-embedding token to each sequence. Train conditionally. Do class-specific attention patterns emerge? Compare unconditional vs. conditional sample grids. Do some classes benefit more from conditioning than others?	Hard

On unexpected results: A result that falsifies your hypothesis is equally valid as one that confirms it. What is not acceptable is ignoring a disconfirming result or re-running until it disappears. Name what happened, offer an explanation, and state what experiment would test that explanation.

Deliverables

Submit all items as a single compressed folder (.zip). Label each document clearly with your name and student ID.

1. **Runnable notebook (.ipynb)** Runnable notebook (.ipynb) —
 - Structured clearly by milestone (use markdown cell headers)
 - All plots must be labeled with axis titles, legends, and a one-sentence caption
 - Must run cleanly top-to-bottom with no dead or error-producing cells
2. **Attention timeline figures** Attention timeline figures —
 - Two figures showing attention weights across generation steps (Milestone 2)
 - Each figure accompanied by a written caption of 3–5 sentences interpreting what is shown
3. **Comparison memo (300 words)** Comparison memo (300 words) —
 - VAE vs. causal transformer across the three axes defined in Milestone 3
 - Use-case recommendations with explicit references to your experimental results
4. **Investigation report (400–600 words)** Investigation report (400–600 words) —
 - Hypothesis stated before results
 - Results with supporting plot(s)
 - Discussion that addresses at least one unexpected finding
 - One open question your data did not resolve
5. **AI usage log** AI usage log —
 - Tabular or prose format, included as an appendix to the notebook or as a separate document
 - For each significant use: the prompt given, what was produced, what you changed, and why
 - Must include at least one instance where the AI output was wrong, incomplete, or misleading

Academic integrity: Using AI tools is not only permitted but expected. The AI usage log is a record of your intellectual process, not a confession. Submitting work without engaging with it — copying AI output without understanding, verifying, or extending it — is a violation of academic integrity regardless of which tool produced the text or code.

Rubric

The rubric is weighted to reflect the shift described in Section 2: implementation carries 30% of the grade; interpretation, comparison, and communication carry the remaining 70%.

Component	Criteria	Points
Causal transformer — correct implementation	No future-token leakage verified; training loss decreasing; next-patch predictions sensible on held-out examples	15
Autoregressive sampling + attention timelines	20 samples generated and displayed; 2 annotated attention-timeline figures with written captions specific to those images	25
VAE comparison — three axes	Both models trained on same dataset; fair comparison on sample quality, training speed, and generation speed; use-case recommendations grounded in results	25
Investigation quality	Hypothesis stated upfront; ≥ 4 data points; unexpected result engaged with honestly; 400–600 word discussion	30
AI usage log	Honest and specific; includes at least one case where AI output was wrong or required correction	5
Total		100

Grading notes

- Attention timeline captions are graded on specificity. Generic statements such as "the model attends to nearby patches" without reference to the actual image or timestep receive partial credit only.
- The VAE comparison is graded on fairness of experimental setup (same dataset, same epochs, same evaluation metric) as well as quality of reasoning.
- Investigation quality is assessed on the strength of the argument, not on whether the hypothesis was correct.

Background reading (optional)

The following papers are not required but may be useful for context and inspiration:

- van den Oord et al. (2016). "Pixel Recurrent Neural Networks." ICML. — the autoregressive generation framework this project extends to patches.
- Chen et al. (2020). "Generative Pretraining from Pixels (iGPT)." ICML. — pixel-level autoregressive generation with transformers.
- Kingma & Welling (2014). "Auto-Encoding Variational Bayes." ICLR. — the VAE baseline you will implement and compare against.